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ORIGINAL ARTICLE

Effect of voluntary hypocapnic hyperventilation on the metabolic response during Wingate anaerobic test

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Abstract
Purpose We evaluated whether hypocapnia achieved through voluntary hyperventilation diminishes the increases in oxygen uptake elicited by short-term (e.g., ~30 s) all-out exercise without affecting exercise performance.
Methods Nine subjects performed 30-s Wingate anaerobic tests (WAnT) in control and hypocapnia trials on separate days in a counterbalanced manner. During the 20-min rest prior to the 30-s WAnT, the subjects in the hypocapnia trial performed voluntary hyperventilation (minute ventilation = 31 L min^{-1}), while the subjects in the control trial continued breathing spontaneously (minute ventilation = 14 L min^{-1}).
Results The hyperventilation in the hypocapnia trial reduced end-tidal CO_2 pressure from 34.8 ± 2.5 mmHg at baseline rest to 19.3 ± 1.0 mmHg immediately before the 30-s WAnT. In the control trial, end-tidal CO_2 pressure at baseline rest (35.9 ± 2.5 mmHg) did not differ from that measured immediately before the 30-s WAnT (35.9 ± 3.3 mmHg). Oxygen uptake during the 30-s WAnT was lower in the hypocapnia than the control trial (1.55 ± 0.52 vs. 1.95 ± 0.44 L min^{-1}), while the postexercise peak blood lactate concentration was higher in the hypocapnia than control trial (10.4 ± 1.9 vs. 9.6 ± 1.9 mmol L^{-1}). In contrast, there was no difference in the 5-s peak (842 ± 111 vs. 850 ± 107 W) or mean (626 ± 74 vs. 639 ± 80 W) power achieved during the 30-s WAnT between the control and hypocapnia trials.
Conclusions These results suggest that during short-period all-out exercise (e.g., 30-s WAnT), hypocapnia induced by voluntary hyperventilation reduces the aerobic metabolic rate without affecting exercise performance. This implies a compensatory elevation in the anaerobic metabolic rate.

Keywords High-intensity exercise · Anaerobic capacity · Respiratory alkalosis · Hypoxia · Training

Abbreviation
WAnT Wingate anaerobic test

Introduction

Arterial CO_2 pressure is usually well maintained within the normocapnic range, as ventilation is tightly regulated to match metabolic needs. However, when ventilation increases to a level beyond that is necessary to meet metabolic needs, arterial CO_2 pressure declines (hypocapnia). This hypocapnia may promote anaerobic energy production during high-intensity exercise, as has been observed with the 30-s Wingate anaerobic test (WAnT) during hypoxic exposure (Calbet et al. 2003). One report has shown that during submaximal exercise, hypocapnia induced by voluntary hyperventilation can reduce muscle blood flow, and thus the oxygen supply, to active muscles (Chin et al. 2010b). In line with that finding, hypocapnia was found to diminish the increases in the aerobic metabolic rate that normally occur at the beginning of submaximal exercise (Chin et al. 2007, 2010a, b; Hayashi et al. 1999; LeBlanc et al. 2002). However, it remains to be seen whether

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hypocapnia also attenuates the aerobic metabolic rate during the WAnT. On the other hand, hypocapnia does not appear to influence WAnT performance (Morrow et al. 1988). This indicates that if hypocapnia reduces the aerobic metabolic rate during the WAnT without a change in performance, there must be a compensatory increase in the anaerobic metabolic rate.

Accordingly, the purpose of the present study was to test the hypothesis that hypocapnia achieved by voluntary hyperventilation lowers the aerobic metabolic rate during 30-s WAnT without affecting exercise performance.

Methods

Ethical approval

The current study was approved by the Human Subjects Committee of the University of Tsukuba and conformed to the provisions of the Declaration of Helsinki. Informed consent was obtained from all participants in the study.

Subjects

Nine college athletes (8 males and 1 female) participated in this study. None of the subjects were smokers or were taking prescription medication. The subjects were 22.8 ± 2.4 (mean ± standard deviation) years of age, 1.73 ± 0.05 m in height, and weighed 65.2 ± 4.1 kg. These athletes included three sprinters, one 800-m runner, one 200-m individual medley swimmer, one soccer defense player, one water polo player, one triathlete and one baseball catcher. We did not consider the menstrual cycle for the one female.

Procedures

Thirty-second WAnTs were carried out in control and hypocapnia trials separated by 5–20 days. The order of the trials was counterbalanced, and the two trials were performed at the same time on each day. An electromagnetic brake bicycle ergometer (Powermax-V2, COMBI, Tokyo, Japan) with toe clips and sprint racing handlebar was used for the 30-s WAnTs, as in our earlier study (Fujii et al. 2012). In both trials, ~10 min after arriving at the laboratory (ambient temperature = ~22 °C, relative humidity = ~50 %), each subject adjusted the seat position on the bicycle ergometer to their individual preference. After all equipment was attached, the subjects rested for 3 min while sitting on the seat of the bicycle ergometer (baseline rest). Each subject then performed a 3-min warm-up at 60 rpm with a 1.5-kp workload, followed by a 5-min rest on the bicycle. Thereafter, in the hypocapnia trial, a 20-min hyperventilation intervention on the bicycle was initiated (hyperventilation rest). During the hyperventilation rest, subjects performed voluntary hyperventilation to reduce end-tidal CO_2 pressure until 10 s before the 30-s WAnT. In the control trial, the subjects continued their spontaneous breathing. In the hypocapnia trial, the subjects maintained a tidal volume of 1000 mL and respiratory frequency of 30 breaths min^{-1}. This respiratory pattern was accomplished using visual feedback from a computer display showing the tidal volume and auditory cues from a metronome for respiratory frequency. This respiratory pattern was selected based on results from an earlier study (Chin et al. 2007), which showed that it produced hypocapnia. We confirmed in our pilot work that this hyperventilation and resultant hypocapnia did not cause severe discomfort such as strong tetany or dizziness. After the 20-min hyperventilation intervention (or spontaneous breathing), each subject commenced the 30-s WAnT, during which he/she was instructed to reach peak revolutions as quickly as possible and to make a full effort until the end of exercise. Pedal resistance was set relative to each subject's body mass (0.1 kp per kg of body mass). After the 30-s WAnT, the subjects dismounted from the bicycle and sat for 10 min in a chair located next to the bicycle ergometer. In both trials, the subjects breathed freely during the 30-s WAnT and 10-min postexercise recovery.

Measurements

Respiratory variables

The subjects breathed from a low-dead space, low-resistance mask that covered the nose and mouth. A pneumotachograph transducer for evaluating respiratory volume was attached to the mask. A gas-sampling tube (the sampling rate was 60 mL min^{-1}) for a mass spectrometer (ARCO-2000, Arco System, Japan), which analyzed respiratory O_2 and CO_2 pressure, was attached to the pneumotachograph transducer. Immediately before the start of measurements, the mass spectrometer was calibrated using reference gases of known concentration, and the pneumotachograph transducer was calibrated using a syringe with a known volume (2 L). Based on the respiratory volume and/or gases measured, the mass spectrometer provided breath-to-breath minute ventilation, tidal volume, respiratory frequency, end-tidal CO_2 pressure, end-tidal O_2 pressure, oxygen uptake and CO_2 output. Air expired during the 30-s WAnT was collected into a Douglas bag to determine oxygen uptake and minute ventilation. After the 10-min postexercise recovery, the internal volume of the Douglas bag was measured using a dry gas meter (DC-5A; Shinagawa; Tokyo, Japan), and the mixed expired O_2 and CO_2 fractions in the Douglas bag were analyzed using the aforementioned mass spectrometer. We used a Douglas bag during the 30-s WAnT because in the pilot work we occasionally found artifacts in the breath-to-breath data from the mass spectrometer collected during the 30-s WAnT. These were the result of the rapid changes in respiration and body movements associated with performing the WAnT.

Power variables during 30-s WAnT

The power output and pedaling rate were fed into a computer (VAIO PCG. 952B, Sony, Japan) every 0.1 s and were averaged every 5 s.

Blood lactate concentration

Blood samples taken from a warmed finger were analyzed using a lactate analyzer (YSI 2300 SPORT; YSI, Ohio, USA) to determine blood lactate concentration.

Heart rate

Heart rate was recorded every 5 s using a heart rate monitor (HR monitor Vantage NV, POLAR, Kempele, Finland).

Rate of perceived exertion

6 to 20 point ratings of perceived exertion (Borg 1975) were recorded immediately after the 30-s WAnT.

Data analysis

We determined peak power from the highest 5-s power and the minimum power from the lowest 5-s power recorded during the 30-s WAnT. Mean power was determined by averaging the power over the entire 30 s of the WAnT. A fatigue index was calculated as the absolute difference between the peak and minimum power expressed as a percent of the peak power. Respiratory and heart rate values during the baseline rest were obtained by averaging measurements made over a 3-min period before the warm-up. Respiratory and heart rate values during the hyperventilation rest were obtained by averaging values recorded over a span from the 14th to the 19th min of the 20-min hyperventilation rest. Peak heart rate was determined from highest 5-s value observed during the 30-s WAnT. Δ heart rate was determined as the peak heart rate minus the heart rate recorded at the hyperventilation rest. The blood lactate concentrations at the baseline and hyperventilation rests were determined from blood samples collected immediately before the warm-up and 2 min before the 30-s WAnT, respectively. The peak blood lactate concentration was determined as the highest value obtained during the 10-min recovery period (i.e., the highest value from among the sample times: 40 s and 2.5, 5.0, 7.5 and 10 min). The Δ blood lactate concentration was calculated as the difference between the peak and hyperventilation rest.

Statistical analysis

Respiratory variables measured before and after the 30-s WAnT were analyzed using two-factor repeated-measures analysis of variance with CO_2 (control or hypocapnia) and time (baseline rest, hyperventilation rest, peak or recovery at 2, 4, 6, 8 and 10 min) as factors. Blood lactate concentrations and heart rates were also analyzed using two-factor repeated-measures analysis of variance with CO_2 (control or hypocapnia) and time (baseline rest, hyperventilation rest or peak) as factors. After determining the main effects, pairwise differences were identified using two-tailed paired t tests with Holm–Bonferroni's correction. Two-tailed paired t tests were also used to compare variables between trials, where applicable. Values of $P \leq 0.05$ were considered significant. Values are represented as the mean ± standard deviation.

Results

Respiratory variables

There was a significant interaction of CO_2 and time for both minute ventilation and end-tidal CO_2 pressure (both $P < 0.001$). As designed, during the hyperventilation rest, minute ventilation was higher (31.4 ± 5.8 vs. 14.5 ± 2.3 L min^{-1}, $P < 0.001$) while end-tidal CO_2 pressure was lower (19.3 ± 1.0 vs. 35.9 ± 3.3 mmHg, $P < 0.001$) in the hypocapnia than control trial (Fig. 1). All subjects in the hypocapnia trial exhibited a reduction in oxygen uptake during the 30-s WAnT as compared to the control trial (1.55 ± 0.52 vs. 1.95 ± 0.44 L min^{-1}, $P < 0.001$, Fig. 2a). On average, the decrease was 21.3 ± 13.1 %. In contrast, there was no difference in minute ventilation between the hypocapnia and control trials during the 30-s WAnT (84.8 ± 53.9 vs. 88.4 ± 41.8, $P = 0.712$, Fig. 2b).

An interaction was evident for end-tidal O_2 pressure ($P < 0.001$). As compared to the control trial, end-tidal O_2 pressure was higher in the hypocapnia trial during the hyperventilation rest (116 ± 5 vs. 136 ± 2 mmHg, $P < 0.001$), but not during the baseline rest (116 ± 5 vs. 115 ± 4 mmHg, $P = 0.867$) or at any time during the recovery (all $P > 0.058$).

Although a main effect of time was detected ($P < 0.001$), no interaction ($P = 0.670$) and no main effect of CO_2 ($P = 0.657$) were found for oxygen uptake. This means oxygen uptake did not differ between the control and hypocapnia trials at the baseline rest (0.50 ± 0.15 vs. 0.48 ± 0.12 L min^{-1}), hyperventilation rest (0.47 ± 0.10

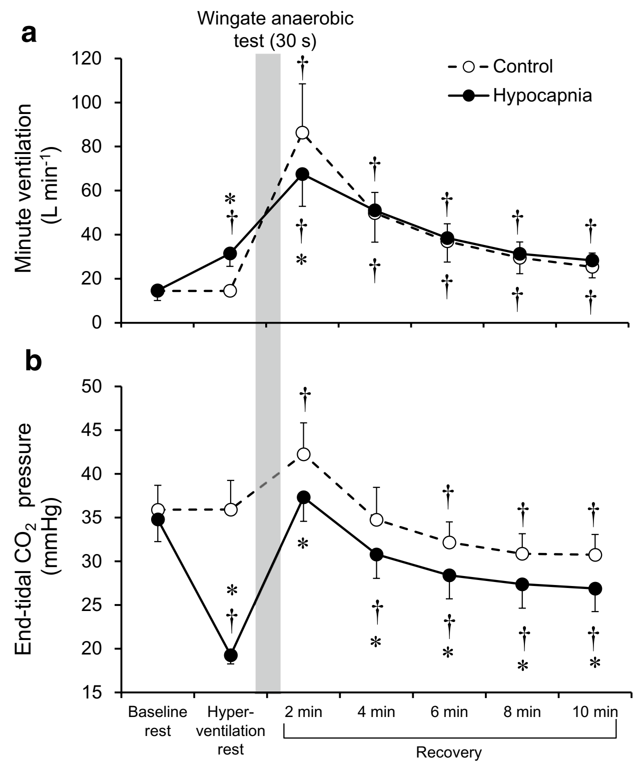

Fig. 1 Time course of changes in minute ventilation (**a**) and end-tidal CO_2 pressure (**b**). *$P < 0.05$ vs. Control. $^{\dagger}P < 0.05$ vs. Baseline rest within a trial

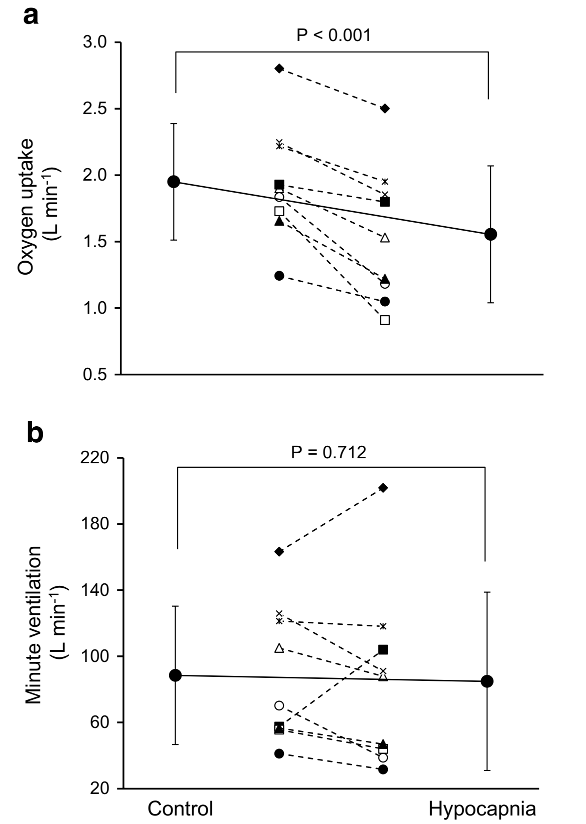

Fig. 2 Oxygen uptake (**a**) and minute ventilation (**b**) measured during the 30-s Wingate anaerobic test. Both individual data and the means with standard deviations are shown

vs. 0.45 ± 0.09 L min^{-1}) or at any time during the recovery. We did detect an interaction for CO_2 output ($P < 0.001$), and post hoc analysis revealed that although there was no difference in CO_2 output between the control and hypocapnia trials at the baseline rest (0.44 ± 0.12 vs. 0.41 ± 0.13 L min^{-1}, $P = 0.590$) or the hyperventilation rest (0.43 ± 0.08 vs. 0.49 ± 0.12 L min^{-1}, $P = 0.125$), it was higher in the control trial than the hypocapnia trial 2 min into the recovery (3.02 ± 0.93 vs. 2.13 ± 0.44 L min^{-1}, $P = 0.008$). Thereafter, no between-trial difference in CO_2 output was measured (all $P > 0.066$).

No interaction ($P = 0.192$), but main effects of CO_2 ($P = 0.010$) and time ($P < 0.001$) were detected for tidal volume. Tidal volume at baseline rest did not differ between the control and hypocapnia trials (0.76 ± 0.14 vs. 0.80 ± 0.22 L, $P = 0.475$). As compared to the control trial, tidal volume was higher in the hypocapnia trial at the hyperventilation rest (0.82 ± 0.19 vs. 1.05 ± 0.19 L, $P = 0.023$). During recovery, tidal volume tended to be higher in the control than the hypocapnia trial at 2, 4 and 6 min, though the difference was not significant (all $P > 0.087$). An interaction was measured for respiratory frequency ($P = 0.015$). Respiratory frequency at the baseline rest was similar in the control and hypocapnia trials (23 ± 6 vs. 22 ± 8 breaths min^{-1}, $P = 0.892$), whereas it was higher at the hyperventilation rest in the hypocapnia trial (20 ± 3 vs. 30 ± 1 breaths min^{-1}, $P < 0.001$). During recovery, respiratory frequency was lower in the control than the hypocapnia trial at 4 min (29 ± 7 vs. 34 ± 7 breaths min^{-1}, $P = 0.003$) and 6 min (27 ± 4 vs. 31 ± 6 breaths min^{-1}, $P = 0.040$), but not at other times (all $P > 0.105$).

Power variables during the 30-s WAnT

Exercise performance by the subjects in the 30-s WAnT was similar in the control and hypocapnia trials, as reflected by the peak (842 ± 111 vs. 850 ± 107 W, $P = 0.596$, Fig. 3a), mean (626 ± 74 vs. 639 ± 80 W, $P = 0.207$, Fig. 3b) and minimum (479 ± 61 vs. 474 ± 65 W, $P = 0.779$) power levels, as well as the fatigue index (43 ± 7 vs. 44 ± 6 %, $P = 0.582$).

Blood lactate concentration

There was a significant interaction with blood lactate concentration ($P = 0.014$). The blood lactate concentration was

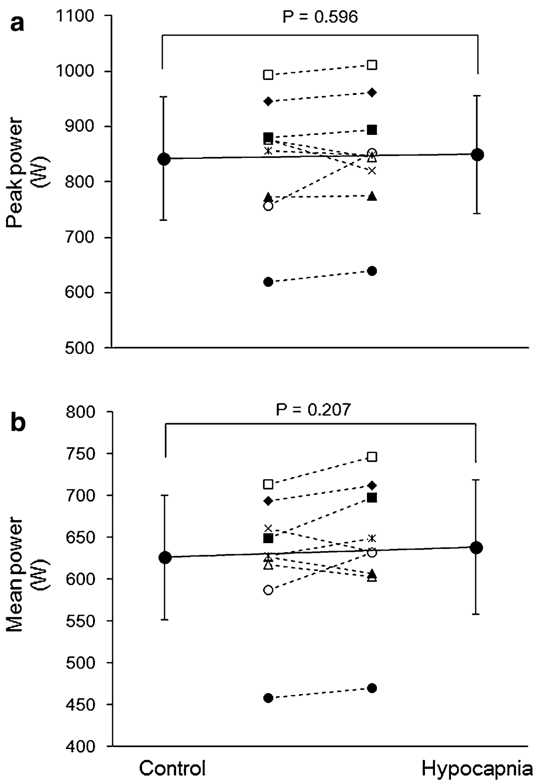

Fig. 3 Peak (**a**) and mean (**b**) power elicited during the 30-s Wingate anaerobic test. Both individual data and means with standard deviations are shown

higher in the hypocapnia than the control trial during the hyperventilation rest (1.4 ± 0.4 vs. 0.9 ± 0.2 mmol L^{-1}, $P = 0.012$) as well as when the concentration was at its peak (10.4 ± 1.9 vs. 9.6 ± 1.9 mmol L^{-1}, $P = 0.020$) (Table 1).

Heart rate

A significant interaction was also detected with heat rate ($P = 0.050$). Heart rate during the hyperventilation rest in the hypocapnia trial tended to be higher than in the control trial (84 ± 18 vs. 75 ± 12 beats·min^{-1}, $P = 0.087$, Table 1). The Δ heart rate in the hypocapnia trial was lower than in the control trial (77 ± 14 vs. 87 ± 12 beats·min^{-1}, $P = 0.029$, Table 1).

Rate of perceived exertion

The rate of perceived exertion immediately after the 30-s WAnT did not differ between the control and hypocapnia trials (17.2 ± 1.6 vs. 17.2 ± 1.4, $P = 1.000$).

Discussion

We evaluated the influence of hypocapnia induced by voluntary hyperventilation on oxygen uptake during short-term all-out exercise (i.e., 30-s WAnT). We found that pre-exercise voluntary hyperventilation-induced hypocapnia reduced oxygen uptake during 30-s WAnTs, though it had no effect on peak or mean power elicited during the test. These results suggest that hypocapnia induced by voluntary hyperventilation attenuates the aerobic metabolic rate during short-term (e.g., ~30 s) all-out exercise without affecting exercise performance. They also imply compensatory enhancement of the anaerobic metabolic rate.

Metabolism

All nine subjects in the present study showed lower oxygen uptake during the 30-s WAnT in the hypocapnia trial

Table 1 Blood lactate concentration and heart rate

	Control	Hypocapnia	*P* value for difference between trials
Blood lactate concentration, mmol L^{-1}			
Baseline rest	0.8 ± 0.2	0.8 ± 0.1	0.497
Hyperventilation rest	0.9 ± 0.2	1.4 ± 0.4†	0.012
Peak	9.6 ± 1.9†	10.4 ± 1.9†	0.020
Δ	8.8 ± 1.9	9.0 ± 1.7	0.512
Heart rate, beats min^{-1}			
Baseline rest	75 ± 10	77 ± 16	0.606
Hyperventilation rest	75 ± 12	84 ± 18†	0.087
Peak	163 ± 9†	162 ± 10†	0.775
Δ	87 ± 12	77 ± 14	0.029

Peak, the highest value obtained during the 10 min of recovery for blood lactate concentration, or during the 30-s WAnT for heart rate. Δ, difference between hyperventilation rest and peak. †, significantly different from baseline rest within a trial ($P < 0.05$)

than the control trial (Fig. 2a), suggesting that hypocapnia induced by voluntary hyperventilation diminishes the increase in the aerobic metabolic rate elicited by short-term all-out exercise. Performance evaluated based on peak and mean power was similar in the two trials (Fig. 3), which means the reduction in oxygen uptake was not due to a reduction in total energy demand, assuming mechanical efficiency is not affected by hypocapnia. Our results further suggest that hypocapnia developed through voluntary hyperventilation enhances anaerobic metabolism, which compensates for the reduced aerobic metabolism, resulting in unchanged short-period all-out exercise performance. Assuming a mechanical efficiency of 20 %, the compensated anaerobic metabolic rate of 0.4 O_2 L min^{-1}, which is estimated based on the difference in oxygen uptake between the control and hypocapnia trials, equals ~30 W. This explains approximately 5 % of the mean power output of 630 W. Enhancement of the anaerobic metabolic rate is also indirectly supported by our finding that peak blood lactate concentrations were higher during postexercise recovery in the hypocapnia than control trial (Table 1).

In contrast to our result, McLellan et al. (1993) observed that oxygen uptake measured during a 45-s WAnT was not lower under hypocapnic hypoxia than under normocapnic hypoxia, and exercise performance did not differ between the two conditions. It may be that the effect of hypocapnia on aerobic metabolism during exercise differs depending on the level of the O_2 partial pressure. In addition, it should be mentioned that the magnitude of hypocapnia developed in the study from McLellan et al. (1993) was much smaller (4 mmHg reduction in end-tidal CO_2 pressure) than in the present study or in other earlier studies (Chin et al. 2010a, b; Chin et al. 2007; Hayashi et al. 1999; LeBlanc et al. 2002), wherein voluntary hyperventilation-mediated hypocapnia diminished increases in oxygen uptake during exercise (10–20 mmHg reduction in end-tidal CO_2 pressure). Hence, there may be a hypocapnic threshold somewhere between a 4 and 10 mmHg reduction in end-tidal CO_2 pressure, at which point the reduction is sufficient to attenuate aerobic metabolism during exercise.

The precise mechanism(s) by which voluntary hyperventilation-induced hypocapnia lowers oxygen uptake during the 30-s WAnT is unclear. That minute ventilation did not differ between the trials (Fig. 2b) means pulmonary ventilation is not a main factor mediating the reduction in oxygen uptake. No association between minute ventilation and oxygen uptake is evident from two subjects who exhibited a more robust increase in minute ventilation in the hypocapnia than the control trial, and showed a reduction rather than an increase in oxygen uptake in the hypocapnic trial (black diamonds and squares in Fig. 2). One possible mechanism is that hypocapnia induces a leftward shift in the oxy-hemoglobin dissociation curve, which would impair O_2 off-loading from hemoglobin and thus reduce the oxygen supply to active muscle, as was suggested by Hayashi et al. (1999). However, this is unlikely in the present study, as Chin et al. (2007) demonstrated that this effect lasts only ~7 min from the start of voluntary hypocapnic hyperventilation. Δ heart rate was smaller in the hypocapnia than the control trial in the present study (Table 1), which is consistent with a previous study in which hypocapnia induced by voluntary hyperventilation blunted the tachycardia response to submaximal exercise (Chin et al. 2010b). It may be that increases in cardiac output during the 30-s WAnT were smaller in the hypocapnia than control trial, reflecting reduced blood flow and thus reduced oxygen supply to active muscles, which would in turn reduce oxygen uptake. In that context, Chin et al. (2010b) showed that the active muscle blood flow response to submaximal exercise was diminished by voluntary hyperventilation-induced hypocapnia. Moreover, another previous study employing submaximal exercise reported that hypocapnia induced by voluntary hyperventilation slowed activation of the mitochondrial pyruvate dehydrogenase complex, which is responsible for oxidizing carbohydrate and thus aerobic metabolism (LeBlanc et al. 2002). This mechanism may also be involved in the reduced oxygen uptake observed in the present study.

Exercise performance

Chin et al. (2007) reported that 20 min of voluntary hyperventilation leading to a 16-mmHg reduction in end-tidal CO_2 pressure increased plasma pH from 7.44 to 7.63 (alkalosis) at rest. As we employed the same hyperventilation procedure, a similar alkalosis was probably induced in the present study. Such alkalosis could potentially improve exercise performance by alleviating acidosis triggered by the strenuous exercise, although whether acidosis contributes to fatigue during intense exercise is still controversial (see other papers for details) (Cairns 2006; Lindinger 2007; Westerblad et al. 2002). As pertains to this, reports show that supramaximal exercise performance lasting ~60 s was improved by alkalosis induced by bicarbonate (pH increases by ~0.10) (Douroudos et al. 2006; Goldfinch et al. 1988; Inbar et al. 1983; McNaughton et al. 1991; McNaughton 1992), but other studies could not confirm this effect (Kindermann et al. 1977; McCartney et al. 1983). Importantly, another study showed that acute alkalosis induced by voluntary hyperventilation does not affect power output measured during a 45-s WAnT (Morrow et al. 1988), which is consistent with our finding in the present study. It may be that, in the present study, any beneficial effect of the acute alkalosis induced by voluntary hyperventilation is offset by concomitant effects such as reduced oxygen uptake.

Ventilation

Arterial CO_2 pressure profoundly affects ventilation by altering the medullary H^+ concentration, which stimulates central chemoreceptors (Ainslie and Duffin 2009; Bruce and Cherniack 1987). Consistent with that notion, we observed that, immediately after the 30-s WAnT (i.e., after 2 min of recovery), end-tidal CO_2 pressure and minute ventilation were both higher in the control than hypocapnia trial (Fig. 1). In addition, lower end-tidal CO_2 pressure before and after the 30-s WAnT in the hypocapnia trial (Fig. 1b) indicates there was less central chemoreflex activation driving ventilation during the 30-s WAnT. Nonetheless, minute ventilation during the 30-s WAnT was similar in the two trials (Fig. 2b). This differs from a study by Ward et al. (1983), who reported that pre-exercise voluntary hyperventilation-induced hypocapnia attenuates an initial increase in minute ventilation occurring during submaximal exercise. Although the precise mechanisms need clarification, since the blood lactate level was higher in the hypocapnia than the control trial (Table 1), ventilatory drive evoked by the muscle metaboreflex (Iellamo et al. 1999; Piepoli et al. 1995) may have also been higher in the hypocapnia trial, compensating for the lower activation of central chemoreflex during 30-s WAnT. Alternatively, an initial increase in ventilation induced by all-out exercise may be less influenced by CO_2.

Practical implications

An emphasis on anaerobic energy systems during high-intensity exercise training would be important for improving those anaerobic energy systems, and in turn performance. Hypoxia suggestively enhances the anaerobic metabolic rate during high-intensity exercises without influencing exercise performance (Calbet et al. 2003; McLellan et al. 1990, 1993; Ogawa et al. 2007; Ogura et al. 2006; Weyand et al. 1999). This implies that hypoxia would improve the impact of high-intensity exercise training on anaerobic energy systems, though its practicality is currently controversial (Fujimaki et al. 1999; Ogita and Tabata 1999; Truijens et al. 2003). Moreover, hypoxic/altitude training requires visiting high-altitude sites or installing a hypoxic chamber, which adds complications to the training. Hypocapnia is readily induced by voluntary hyperventilation, and our results suggest that hypocapnia amplifies the anaerobic metabolic rate during short-term all-out exercise. Accordingly, we propose future use of a new training protocol featuring "hypocapnic training," which is simpler than hypoxic/altitude training and may be effective for developing anaerobic energy systems. To develop a practical hypocapnic training protocol, further studies will be needed to address as yet unanswered questions, such as what is the level of hypocapnia and minimal duration of hyperventilation required to sufficiently burden the anaerobic energy system while exerting minimal side effects such as dizziness and/or tetany. Furthermore, although the one female subject showed a pattern of responses to hypocapnia that was similar to the males, given the limited sample size and the absence of a control for menstruation, we can draw no conclusions with respect to sex differences or the influence of the menstruation cycle. Clarifying the issues specific to females in the future will be necessary for a full understanding of hypocapnic responses.

Conclusions

Our results suggest that voluntary hyperventilation-induced hypocapnia prior to exercise reduces the aerobic metabolic rate during short-term all-out exercise without influencing exercise performance. This implies that hypocapnia-induced increases the anaerobic metabolic rate.

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Conflict of interest None.

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